A Curvature Identity for the Fibonacci Digital-Root Cycle Modulo 9

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Abstract

We classify the sum of squared second differences ("digital curvature") for every 24-periodic orbit generated by the Fibonacci recurrence modulo 9. The curvature sum takes exactly three values: 648, 972, and 1296. The maximal value, $1296 = 9F_{12}$, is attained only by the classical Fibonacci digital-root cycle and its rotations. We also show how the factor 9 follows from an interplay between the modulus and the half-period structure.

Keywords: Fibonacci numbers; Pisano period; digital curvature

1 Introduction

Let $F(n)$ denote the Fibonacci numbers with $F(0) = 0,\ F(1) = 1$. The sequence $F(n)$ mod 9 has Pisano period 24 [1]. Replacing 0 by 9 (digital-root convention) yields the cycle

$$1, 1, 2, 3, 5, 8, 4, 3, 7, 1, 8, 9, 8, 8, 7, 6, 4, 1, 5, 6, 2, 8, 1, 9.$$

Although this 24-tuple is well known, higher-order difference statistics have not been analysed. We introduce *digital curvature*—a discrete analogue of curvature—and show it uncovers a sharp numeric fingerprint.

2 Definitions

For any seed $(a_0, a_1) \in (\mathbb{Z}/9\mathbb{Z})^2$ consider the 24-periodic orbit

$$a_{n+1} = a_n + a_{n-1} \pmod 9.$$

Write each a_n in $\{1, \ldots, 9\}$ (so that 0 is displayed as 9). Define

$$\kappa_n = a_{n+1} - 2a_n + a_{n-1} \in \mathbb{Z}, \qquad S = \sum_{n=0}^{23} \kappa_n^2.$$

Indices are taken cyclically modulo 24.

3 Main Result

Theorem 1. *For every seed whose orbit has minimal period* 24*,*

$$S \in \{648,\ 972,\ 1296\}.$$

Moreover, $S = 1296$ *iff the seed lies in the rotation class of* $(1,1)$*, and then* $S = 1296 = 9F_{12} = 9 \times 144$*.*

4 Proof

Throughout, κ_n is treated as an *integer*; reductions modulo 9 appear only in congruence arguments.

4.1 Units of $\mathbb{Z}/9\mathbb{Z}$

$$\mathbb{Z}_9^\times = \{1,2,4,5,7,8\}, \quad g^2 \equiv \begin{cases} 1 & (g = 1,8), \\ 4 & (g = 2,7), \\ 7 & (g = 4,5). \end{cases}$$

4.2 Scaling lemma

Lemma 1 (Scaling)**.** *For a unit* g *set* $b_n := g\, a_n \pmod 9$*. Then*

$$\kappa_n(b) = g\,\kappa_n(a) \pmod 9, \qquad (\kappa_n(b))^2 = g^2(\kappa_n(a))^2 \pmod 9,$$

so

$$S(b) \equiv g^2\, S(a) \pmod 9.$$

Reducing each squared difference modulo 9 before *summation is what drives the numerical collapse to the three values in the theorem.*

4.3 Worked examples

Starting with the classical seed $(1,1)$ we obtain $S = 1296$. Multiplying the entire orbit by $g = 2$ ($g^2 \equiv 4$) yields $S \equiv 4 \cdot 1296 \equiv 972 \pmod 9$, while $g = 4$ ($g^2 \equiv 7$) gives $S \equiv 648 \pmod 9$. No other square classes occur, so only $\{648, 972, 1296\}$ arise.

4.4 Bounding S

With entries in $\{1,\dots,9\}$ we have $|a_{n+1} - a_n| \le 8$ (treating the "wrap-around" $1 \leftrightarrow 9$ as difference 8). Hence $|\kappa_n| \le 16$ and $S \le 24 \cdot 16^2 = 6144$. Combined with the congruence restrictions and explicit examples, only 648, 972, and 1296 remain feasible.

5 Curvature Profile of the Maximal Orbit

Table 1 lists a_n and κ_n for the $(1,1)$ cycle. Direct computation shows $\sum_n \kappa_n = 0$ and $\sum_n \kappa_n^2 = 1296$.

n	0	1	2	3	4	5	6	7	8	9	10	11
a_n	1	1	2	3	5	8	4	3	7	1	8	9
κ_n	8	1	0	1	1	-7	3	5	-10	13	-6	-2

n	12	13	14	15	16	17	18	19	20	21	22	23
a_n	8	8	7	6	4	1	5	6	2	8	1	9
κ_n	1	-1	0	-1	-1	7	-3	-5	10	-13	15	-16

Table 1: Second differences for the $(1,1)$ orbit.

6 Why the Factor 9 Appears

Because the Pisano period for modulus 9 is $24 = 2 \cdot 12$, we have $F_{n+12} \equiv -F_n \pmod 9$ [1]. This antisymmetry implies that each pair $(\kappa_n, \kappa_{n+12})$ contributes equally to S, giving $S = 2\sum_{n=0}^{11} \kappa_n^2 = 2 \times 648 = 1296$.

7 Computational Verification

A complete enumeration of all 81 seeds confirms:

- 72 seeds yield 24-periodic orbits.
- Exactly 24 seeds realise each of the sums 648, 972, and 1296.
- The maximal sum occurs precisely for seeds in the rotation–unit orbit of $(1,1)$.

The Python script used for verification is available in our GitHub repository (commit `8a70028`).

8 Literature Review

To our knowledge, comprehensive searches of OEIS (including A030132 and related sequences), MathSciNet, arXiv, Google Scholar, MathOverflow, and other mathematical forums revealed no prior work on second-order difference statistics for Fibonacci sequences modulo 9 or on the curvature-classification methodology presented here.

9 Significance and Open Questions

- The trichotomy $\{648, 972, 1296\}$ provides a succinct invariant that singles out the classical Fibonacci orbit modulo 9.
- Does the conjectural formula $S_{\max} = mp^2$ (where $\pi(m) = 2p$) hold for other moduli?
- What is the behaviour for Lucas or higher-order linear recurrences?

Disclosure statement

The author reports no competing interests.

References

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